KL UNIVERSITY | DDCA | BATCH 13

Design and Simulation of a Digital Access Verification System Using XOR / XNOR Logic

Team members

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Section: _____

Introduction and Objective

Digital access verification system: a circuit that checks whether an entered code matches a stored (authorized) code, and grants or denies access. The same logic is used in keypad locks, ID card readers and PIN checks.

Objective

- Build a 4-bit code checker using XNOR and AND gates
- Compare each entered bit with the stored bit
- Show Access Granted or Denied on LEDs
- Simulate and verify the design in Logisim

Input: user enters a 4-bit code

Compare: XNOR checks each bit

Decide: AND gives grant / deny

Theory and Logic Circuit

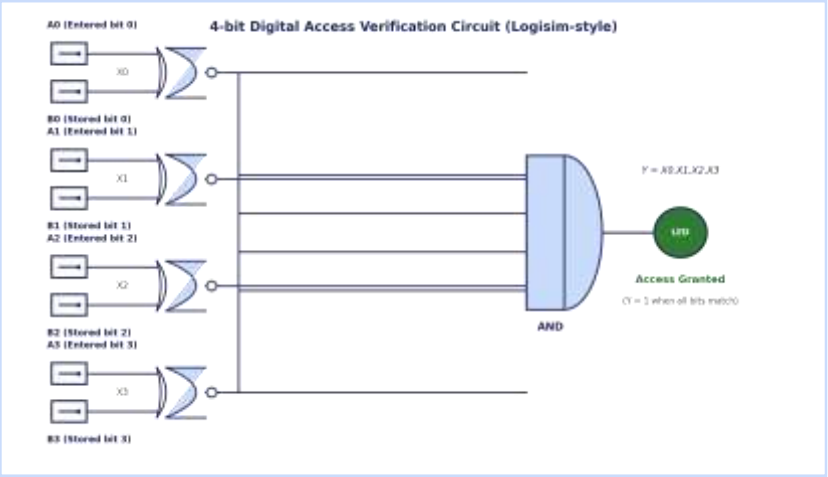
XOR: Y = A ⊕ B

A	B	Y
0	0	0
0	1	**1**
1	0	**1**
1	1	0

XNOR: Y = (A ⊕ B)'

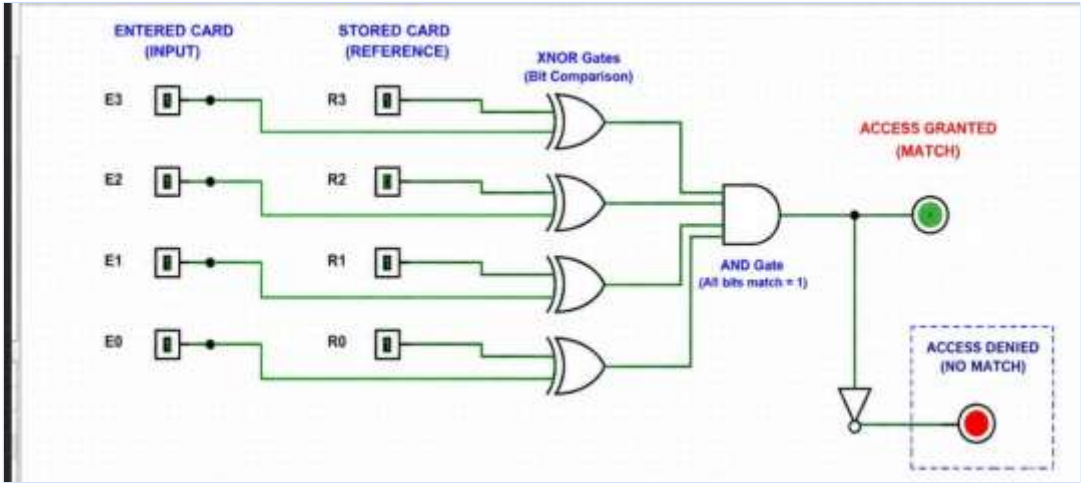
A	B	Y
0	0	**1**
0	1	0
1	0	0
1	1	**1**

XOR gives 1 when the bits differ. XNOR gives 1 when they match, so XNOR is the bit comparator. An AND gate combines the four XNOR outputs.



Logic circuit: 4 XNOR gates (one per bit) feed one AND gate

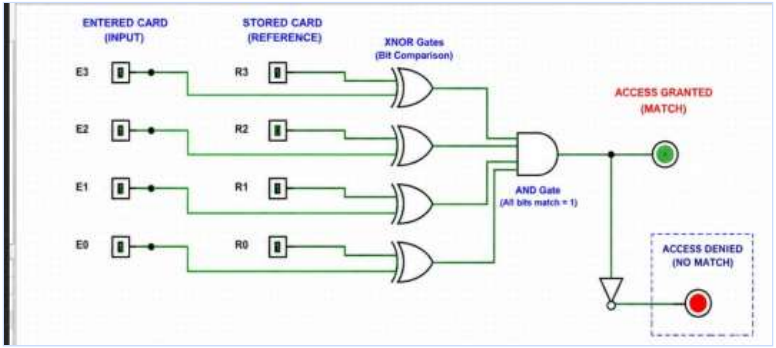
Circuit Design in Logisim



Components: 8 input switches, 4 XNOR gates, 1 AND gate, 1 NOT gate, 2 LEDs

1. Set entered bits E3–E0 and stored bits R3–R0 with switches
2. Each XNOR compares one pair: 1 = match
3. AND outputs 1 only if all four bits match
4. Green LED = granted; NOT gate drives red LED = denied

Simulation Results



Logisim run: entered 1111, stored 1011 (bit 2 differs), so the red LED shows Access Denied

Entered	Stored	XNOR	AND	Result
1011	1011	1111	1	GRANTED
1111	1011	1011	0	DENIED
1010	1011	1110	0	DENIED
0000	1011	0100	0	DENIED

Result: Y = X0·X1·X2·X3 is 1 only when all four bits match, so only the correct code turns the green LED on.

Conclusion and Applications

Outcome

- 4-bit access checker built with XNOR and AND gates
- Verified with truth tables
- Simulated successfully in Logisim

Applications

- Door locks and keypad entry
- ATM PIN checks
- RFID / ID access panels
- Vehicle immobilizers

Possible improvements

- Store the code in a register so it can change
- Add encryption
- Lock after 3 wrong tries
- Extend to 8 or 16 bits

Takeaway: XNOR compares, AND decides. A simple, low-cost access checker.